NASA Space to Soil Challenge: Call for Submissions and Self-Nominations for Judges

Rapid advances in commercial space, artificial intelligence, and edge computing are transforming what is possible for Earth observation. By pushing more intelligence onboard, missions can move from passively collecting data to actively interpreting and responding to changing surface conditions in near-real time, enabling more targeted observations and dramatically improving the value of data returned to the ground. Within this context, land-focused applications such as regenerative agriculture, sustainable forestry, and broader land resilience efforts stand to benefit enormously from satellites that can adapt what, when, and how they sense based on dynamic environmental signals and algorithmic insight rather than fixed schedules or static acquisition plans.

NASA Earth Science Technology Office (ESTO) invites participants to design small satellite (SmallSat) mission concepts that leverage adaptive sensing and onboard processing to enhance regenerative agriculture, forestry, or a similar land resilience objective. Participants must work within onboard power, compute, and bandwidth constraints characteristic of SmallSat missions, focusing on how to orchestrate existing land observation algorithms into an efficient, responsive onboard intelligence layer. Both hardware-oriented and software-oriented solutions—or combinations of the two—are encouraged.

In Phase One, participants will submit a short 5-page white paper and a 2 to 3-minute video of their idea, along with software code (Software) or schematics (Hardware) supporting the Key Capabilities of their solution. Submissions will be evaluated per challenge Judge Criteria. Approximately 3 weeks after the submission deadline, up to 10 Finalists will be selected to present their ideas to a panel of judges at a live Pitch Event in June 2026.

In Phase Two, up to 10 Finalists will be invited to an in-person Pitch Event. Up to 3 Winners from the Pitch Event will receive a $100,000 prize each and be invited to the challenge Incubator Program. Up to 2 Pitch Event Runners-Up will also receive $25,000 each.

In Phase Three, Winners will be invited to the challenge Incubator Program consisting of 10-12 weeks of content. Additionally, winners will be contacted 12 months after Pitch Event completion for follow up surveys on further challenge research development and implementation.

Learn more and register for the NASA Space to Soil Challenge at https://nasa-space-to-soil.org.

Not Planning to Compete in the Challenge, but Interested in Being a Judge?

NASA also is recruiting venture and commercial experts to participate as volunteers in the judging process. Venture experts must be U.S. based and will be required to have a minimum of two or more of the following:

1. Subject matter expertise in small satellite or onboard processing, with particular focus on entrepreneurship.

2. Former experience as a founder of a cutting-edge or disruptive startup with a dual-use business model.
3. Former experience winning SBIR/STTR awards and commercializing the supported technologies.
4. Accreditation and recent experience investing in cutting-edge or disruptive startups.
5. Expertise with venture building from pre-seed through growth stage as a co-founder or early employee.

Self-nominations are requested. The final judging panel will be selected to ensure programmatic balance. Volunteer judges will be required to commit 10-15 hours to the evaluation process in June 2026 as well as participate in a training call in May 2026.

Send an email to info@NASA-Space-to-Soil.org with a subject line to read "NASA Space to Soil Challenge Judge Nomination" if you are interested in participating in the judging process.

Point of Contact
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